

Spatial Outliers And Neighbor Anomalies: A Method For Investigative, Non-Certifying Election Forensics

Gio Della-Libera

Draft — June 28, 2026

Abstract

Neighboring reporting units often resemble one another because they share local geography, administration, demography, and political context. A reporting unit that diverges sharply from its geographic neighbors can therefore be a useful candidate for review. It is not, by itself, evidence of fraud, proof of error, or certification evidence. This W.05 method paper instantiates the W.01 forensic report contract for spatial outliers: neighbors are defined by shared boundaries, features are derived from hash-bound RCOUNT reporting units and geography, scores may come from local Moran's I/LISA statistics, spatial-lag residuals, or neighbor-difference z-scores, and every output remains an investigation queue. The paper defines the method, input contract, intended fixture roles, and false-positive boundary. It does not claim a shipped forensics crate, function, fixture, or test.

1 Introduction

Forensic spatial-outlier analysis can prioritize investigation. It cannot prove fraud, certify correctness, or replace a risk-limiting audit. A high score means that a reporting unit differs from nearby units enough to deserve explanation; it is not itself the explanation.

The intuition is simple. Neighboring reporting units usually resemble each other because they share local geography, administrative practice, media markets, commuting patterns, and demographic context. When one unit diverges sharply from its geographic neighbors in turnout, vote share, residuals, or audit-discrepancy features, the unit is a candidate for review. The review may find a benign reason, a data problem, a boundary artifact, or nothing at all.

This paper is a method and framework paper in the W-series. W.01 defines the shared forensic report contract for investigative, non-certifying anomaly analytics. W.05 instantiates that contract for spatial neighbor anomalies. It describes the statistics, inputs, report fields, and intended fixtures. It does not claim that a W-series forensics crate, fixture suite, or production report already exists.

2 Method

The spatial-outlier method starts with reporting units, unit-level features, and a neighbor graph. Two units are neighbors when their geographic polygons share a boundary segment. The same general reporting-unit adjacency discipline used by the redistricting side of the repository is the right kind of input here, but this paper does not claim a forensics implementation over that machinery.

Let x_i be the feature for unit i , such as turnout rate, candidate share, a model residual, or an audit-discrepancy rate. Let $N(i)$ be the set of neighbors that share a boundary with i , and let w_{ij} be normalized neighbor weights. A spatial-lag comparison computes

$$\ell_i = \sum_{j \in N(i)} w_{ij} x_j,$$

and scores the gap between x_i and ℓ_i . A simple neighbor-difference z-score uses

$$z_i = \frac{x_i - \mu_{N(i)}}{\sigma_{N(i)}},$$

with a stated rule for small or zero-variance neighbor sets. A local spatial-lag residual first fits a baseline model and then compares each unit’s residual to the residuals around it.

Local Moran’s I and related LISA statistics ask whether the value at one unit is unusual relative to the local spatial pattern. They can flag high-low and low-high outliers, as well as clusters where adjacent units jointly depart from the wider field. A W.05 report may therefore identify both local outlier units and spatial-outlier clusters. The statistic used must be recorded in `model_id`, and the variables used must be recorded in `feature_set`.

The method must also preserve the ordinary geography of elections. Urban-rural edges, college towns, military bases, reservation boundaries, retirement communities, precinct splits, and county or municipal borders can create real spatial discontinuities. A good W.05 report flags the discontinuity for review and carries the caveat that sharp neighbor differences can be legitimate.

3 RCOUNT Inputs And Report Contract

The required inputs are RCOUNT reporting units, normalized election features, and geography sufficient to define shared-boundary adjacency. The input package also supplies source hashes, reporting-unit lineage, and any feature provenance needed to reproduce the score. If a jurisdiction supplies geography at a different level than results, the lineage and aggregation rule are part of the feature set, not an informal preprocessing step.

W.05 instantiates the W.01 forensic report contract with spatial-outlier semantics:

Field	W.05 meaning
<code>method_id</code>	Registered W.05 spatial-outlier analytic.
<code>feature_set</code>	Unit features scored, such as turnout, vote share, residuals, or discrepancy rates.
<code>baseline_population</code>	The neighbor set, plus any wider comparison population or time window.
<code>model_id</code>	<code>spatial-outlier</code> , with the specific statistic or residual model named.
<code>unit_id</code>	Reporting unit being scored.
<code>score</code>	Local outlier statistic, residual, or neighbor-difference score.
<code>p_value_or_rank</code>	Optional calibration, permutation p-value, percentile, or queue rank.
<code>investigation_note</code>	Machine-readable caveat, including edge effects or known boundary explanations.

The report metadata binds the output to source package hashes, geography hashes, feature definitions, adjacency construction rules, and model parameters. That binding is essential: a second analyst with the same RCOUNT package and the same registered method should be able to recompute the same investigation queue. The contract is defined in W.01; W.05 only specializes it for neighbor anomalies.

4 Fixture Pattern

The intended W.05 fixtures follow the four-role W-series pattern. They are specified here as design requirements, not as shipped tests.

Fixture	Role
No-anomaly smooth field	A synthetic package where neighboring units vary smoothly. The method should return low scores or an empty low-priority queue.
Injected spatial outlier	A package with one planted unit that sharply departs from its neighbors. The method should surface that unit near the top of the queue.
Innocent discontinuity	A package with a real boundary discontinuity that looks unusual but is known to be benign, exercising the false-positive caveat.
Explained demographic edge	A boundary fixture where a sharp neighbor difference is fully explained by a known demographic edge, such as an urban-rural or college-town boundary.

These fixtures test two behaviors at once. The method should notice a planted local anomaly, and the report should refuse to convert an explainable spatial edge into a fraud claim. The false-positive fixtures are therefore not secondary; they are the part of the method that protects the investigative boundary.

5 Boundaries

Spatial outliers reflect real geography as often as they reflect data problems. Neighboring units can differ because they sit on opposite sides of a city line, county border, campus

boundary, reservation boundary, commuting corridor, or housing-market break. A W.05 score is therefore descriptive. It says that a unit is unusual relative to its local spatial baseline; it does not say why.

The neighbor definition drives the result. Shared-boundary adjacency, rook versus queen-style contact rules, minimum boundary length, islands, water crossings, enclaves, precinct splits, and missing geometries can all change the neighbor set. Edge units also have fewer neighbors and can receive unstable scores. The report must carry enough adjacency metadata and investigation notes for a reviewer to see those choices.

The claim boundary is the same one stated at the beginning. Spatial forensic analytics can prioritize investigation. They cannot prove fraud, certify an election, or replace a risk-limiting audit. Any public use of W.05 should keep its transcript separate from V-series certification evidence and should present high scores as questions for review, not answers.

6 Conclusion

W.05 defines a spatial-outlier method for the W-series: build a shared-boundary neighbor graph over RCOUNT reporting units, compare each unit to its local spatial baseline, record the statistic and caveats in the W.01 report contract, and return an investigation queue. The useful signal is local surprise, not proof.

The method is deliberately modest. It names established statistics, defines the RCOUNT inputs they require, specifies the intended four-role fixture pattern, and states the false-positive boundary. Implementation, fixtures, and any crate placement remain downstream work. Until then, W.05 is a contract for how spatial neighbor anomalies should be reported if they are added to the RCOUNT-adjacent forensics track.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Spatial outliers and neighbor anomalies for election forensics. Technical report, Apportionment Research, 2026b.